Rapid Revision Summary

Last-minute review · condensed from 17 mark schemes and 16 papers, Jan 2019 – Jun 2026

THE PAPER

2 hours · 110 marks · 9–11 questions · ≈ 1 min per mark. Every paper: MCQs on labelled diagrams, a graph to plot, calculations, a 6-mark investigation, 4–6 mark data evaluations.

TOP 12 MARK-LOSERS

- "Amount" — say **volume, mass, concentration, number**.
- Investigation without CORMS points.
- Graph: wrong scale, extrapolated to 0, bar chart (loses L).
- Enzymes "killed" — say **denatured, active site**.
- Osmosis without **water potential**.
- "Faster collisions" — say **more collisions, more E–S complexes**.
- Word equation when symbol equation asked.
- Mineral ion without a matching function.
- Figures quoted with no comparison.
- "Breathing" for respiration; energy "made".
- Pyramid drawn when a food chain asked.
- Unqualified "pollution", "harmful", "climate change".

INVESTIGATIONS AND GRAPHS

CORMS (6)

C change IV (≥5 values) · **O** same species/age/size · **R** repeat + mean · **M1** measure DV with apparatus · **M2** after stated time · **S** two named controls (temperature, volume, concentration, light).

GRAPH (S L A P K)

Scale linear, half grid · Lines through points (or bars if asked) · Axes right way, labels, units · Plots ±½ square · Key for two lines.

- Describe: trend + values + units + levelling off; then explain.
- Median beats mean when data are skewed.
- Reliability = repeats + mean; validity = controls.

CALCULATIONS

Type	Example
% change	(0.08 − 0.04) ÷ 0.04 × 100 = 100%
% of total	356 ÷ 1687 × 100 = 21.1%
ratio	85 147 : 7206 = 11.8 : 1
magnification	43 000 µm ÷ 65 = 660 µm
SA:V	3 cm cube 54 : 27 = 2 : 1
rate	20 cm ÷ 30 s = 0.67 cm/s
standard form	2.5 ÷ 15 000 = 1.7×10^{-4}
time	6250 s ÷ 60 = 104 min

ORGANISMS AND CELLS

Group	Features
plants	chloroplasts, cellulose wall, starch
animals	no wall, glycogen, nerves
fungi	hyphae, chitin, saprotrophic
bacteria	no nucleus, plasmids, loop of DNA
protoctists	single cell with nucleus (Amoeba, Plasmodium)
viruses	protein coat + DNA/RNA, no organelles

- Life: nutrition, respiration, excretion, response, movement, homeostasis, reproduction, growth.
- Mitochondria: respiration · ribosomes: protein · chloroplasts: photosynthesis (palisade).
- Tissue: similar cells, same function.

MOVEMENT AND ENZYMES

DEFINITIONS

Diffusion: high → low concentration. **Osmosis:** water, high → low water potential, partially permeable membrane. **Active transport:** against gradient, ATP.

ENZYMES

Warmer → more KE → more collisions → more E–S complexes. Above optimum → denatured → active site changes → substrate can't fit. More enzyme → faster until substrate limits.

Potato: mass lost in high sucrose, gained in low; no change = equal water potential; blot dry. (2.3 − 2.2) ÷ 2.2 = 4.5%.

PLANT NUTRITION

- $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (light → chemical energy).
- Limiting factors: light, CO_2, temperature.
- Starch test: destarch in dark 24–48 h → boil → ethanol (water bath) → rinse → iodine → blue-black.

- Controls: foil (light), soda lime (CO_2), variegated leaf (chlorophyll).
- Nitrate → protein · Mg → chlorophyll · phosphate → DNA/ATP · K → enzymes.
- Waterlogged soil → no O_2 → no respiration → no active transport.

HUMAN NUTRITION

Test	Positive
iodine	brown → blue-black
Benedict's + heat	blue → brick red
biuret	blue → purple
ethanol + water	milky emulsion

- A vision · C scurvy · D bones · Ca bones · Fe haemoglobin · fibre peristalsis.
- Bile: emulsifies fat, neutralises acid. Villi: large SA.
- Excess carbohydrate → fat → obesity, diabetes, CHD.

RESPIRATION AND BREATHING

- $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ ATP). Muscle anaerobic → lactic acid; yeast → ethanol + CO_2.
- Respirometer: KOH absorbs CO_2; liquid moves towards organism.
- Exhaled air turns limewater cloudy faster.

INHALATION (4–6)

Diaphragm contracts and flattens; intercostals contract; ribs up and out; volume ↑; pressure ↓; air in. Trachea → bronchi → bronchioles → alveoli.

- Damaged lungs: less SA → less O_2 → less aerobic respiration → lactic acid.
- After exercise: high breathing removes lactic acid (oxygen debt).

TRANSPORT AND COORDINATION

- Plasma: glucose, amino acids, CO_2, urea, hormones. RBC: haemoglobin, no nucleus.
- Antibodies specific to antigens; phagocytes engulf.
- CHD risks: genes, blood pressure, fat, smoking, stress, inactivity, obesity.
- Xylem: water + minerals · phloem: sucrose + amino acids.
- Nerves: electrical, fast, specific, short. Hormones: chemical, slow, widespread, long.
- Roots: towards gravity, away from light; shoots opposite.

REPRODUCTION AND INHERITANCE

- Anther (pollen) · stigma (pollen lands) · ovary → fruit · ovule → seed. Wind: exposed anthers, feathery stigma.
- Asexual: clones, fast, no variation. Sexual: variation, new varieties.
- Germination: water, O_2, warmth. Salt → low water potential → less water in.
- Fertilisation: gamete nuclei **fuse** (not "meet").
- Sperm: acrosome, mitochondria, haploid nucleus.
- Aa × Aa → 3 : 1; affected boy = ¼ × ½ = 1/8.
- Natural selection: mutation → variation → survive → reproduce → pass allele.

ECOLOGY AND HUMAN IMPACT

- Chain arrows → eater. Energy lost: respiration, movement, excretion, faeces.
- Decomposers: enzymes, digest dead matter, respire → CO_2.
- Quadrats: random, count, repeat, mean × area.
- Biological control: specific, no reapplying, no bioaccumulation, no resistance; may become a pest.
- Greenhouse: CO_2, methane → heat trapped → global warming.

BIOTECHNOLOGY AND DISEASE

- Yoghurt: pasteurise → Lactobacillus → lactose → lactic acid.
- GM: restriction enzyme cuts gene + plasmid (sticky ends) → ligase joins → vector inserts.
- GM crops: fewer pesticides vs cross-pollination, pollinators, resistance, patents.
- Plasmodium → malaria · TMV → mosaic leaves.
- Trials: sample size, duration, chance, other lifestyle factors.